

Multi-Face Emotion Detection Using CNN Tiny Face Detector

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Abstract: This study presents a real-time multi-face emotion detection system designed for classroom learning environments. The system integrates a CNN-based Tiny Face Detector for multi-scale face localization with a convolutional neural network for emotion classification into seven categories: angry, disgust, fear, happy, sad, surprise, and neutral. Experimental evaluation was conducted using classroom video streams under varying lighting conditions, face orientations, occlusions, and multiple face appearances per frame. The system achieved stable real-time performance with frame rates ranging from **10–20 FPS**, depending on the number of detected faces. Average classification confidence reached up to **91%** for expressive emotions such as happy and surprise, while subtle emotions such as fear and disgust showed lower confidence, decreasing by approximately **15–20%**. Overall classification accuracy remained above **70%** when emotion outputs were aggregated over multiple faces and time windows. The emotion distribution analysis enables objective assessment of classroom emotional dynamics, demonstrating the feasibility of deploying lightweight CNN-based emotion recognition systems for continuous educational environment monitoring.

Keywords: facial emotion recognition; multi-face detection; tiny face detector; CNN; classroom monitoring

Abstrak: Penelitian ini mengusulkan sistem deteksi emosi multi-wajah secara real-time yang dirancang untuk lingkungan pembelajaran di kelas. Sistem mengintegrasikan Tiny Face Detector berbasis CNN untuk pelokalan wajah multi-skala dengan jaringan saraf konvolusional untuk klasifikasi tujuh emosi, yaitu marah, jijik, takut, senang, sedih, terkejut, dan netral. Evaluasi eksperimen dilakukan menggunakan video kelas dengan variasi pencahayaan, orientasi wajah, oklusi parsial, serta jumlah wajah dalam satu frame. Sistem menunjukkan kinerja real-time yang stabil dengan kecepatan pemrosesan antara **10–20 FPS**, bergantung pada jumlah wajah terdeteksi. Rata-rata tingkat kepercayaan klasifikasi mencapai hingga **91%** untuk emosi ekspresif seperti senang dan terkejut, sementara emosi dengan ciri halus seperti takut dan jijik mengalami penurunan kepercayaan sekitar **15–20%**. Akurasi keseluruhan sistem berada di atas **70%** ketika hasil emosi diagregasi berdasarkan banyak wajah dan interval waktu. Analisis distribusi emosi memungkinkan evaluasi objektif suasana kelas dan menunjukkan potensi sistem sebagai alat pemantauan lingkungan pembelajaran berbasis kecerdasan buatan.

Kata kunci: pengenalan emosi wajah; deteksi multi-wajah; tiny face detector; CNN; pemantauan kelas

INTRODUCTION

Emotional conditions play an important role in influencing human behavior, interaction, and performance, particularly in learning environments. In a campus setting, students' emotions can reflect engagement, motivation, stress, and overall learning atmosphere. Conventional methods for evaluating learning conditions, such as questionnaires and direct observation, are often subjective, time-consuming, and limited in real-time analysis. Therefore, an automatic and objective approach to emotion detection is required to support more accurate evaluation of learning environments[1], [2]

Recent advances in computer vision and deep learning have enabled the development of facial emotion recognition systems using Convolutional Neural Networks (CNN). CNN-based methods are capable of extracting high-level facial features and have demonstrated strong performance in emotion classification tasks[3], [4], [5], [6]. However, many existing approaches focus only on single-face detection and are less effective in crowded environments such as classrooms, where multiple faces appear simultaneously with varying scales and lighting conditions.

To address this limitation, multi-face detection techniques have gained increasing attention. Tiny Face Detector is a CNN-based face detection algorithm designed to detect faces of various sizes, including small and distant faces[7],[8]. This method is particularly suitable for real-world scenarios where faces may not always appear clearly or at close range. By integrating Tiny Face Detector with a CNN-based emotion classification model, a robust multi-face emotion detection system can be developed.

Several previous studies have explored facial emotion recognition using CNN architectures and face detection techniques[6], [9], [10], [11]. Research has shown that CNN models outperform traditional machine learning methods in terms of accuracy and robustness. Other studies have highlighted the effectiveness of Tiny Face Detector in handling scale variations and dense face distributions. Nevertheless, limited research has combined Tiny Face Detector with CNN-based emotion recognition specifically for evaluating learning atmospheres in campus environments.

Based on this gap, this study aims to develop a multi-face emotion detection system using CNN and Tiny Face Detector to analyze emotional conditions in campus learning environments. The proposed system is expected to provide real-time emotion analysis, support objective evaluation of learning atmospheres, and contribute to the development of intelligent educational monitoring systems.

METHOD

This study employs an experimental research methodology to design, implement, and evaluate a multi-face emotion detection system for campus learning environments. The proposed method consists of four sequential stages: data acquisition, multi-face detection, emotion classification, and visualization. The workflow is designed to ensure reproducibility and objective performance evaluation.

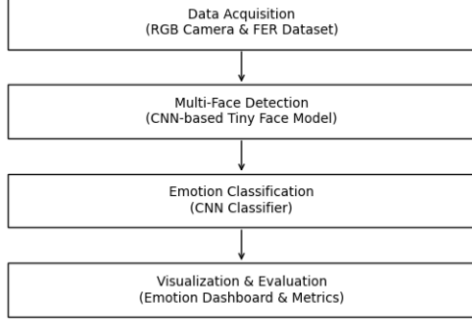


Image 1: Research Stages

Facial image and video data are collected from classroom environments using standard RGB cameras under varying lighting conditions. To improve generalization, the system utilizes the publicly available FER-2013 dataset combined with limited campus-collected samples, following best practices in emotion recognition research[12]. FER-2013 and similar datasets have been widely used to train and benchmark CNN emotion models due to their diversity and labeled emotion classes.

Face localization is performed using a multi-scale CNN-based detection algorithm. The detector used in this study builds upon the Tiny Face Detector concept by aggregating multi-scale features and contextual attention, enabling robust detection of faces of varying sizes in crowded scenes. The face detection process can be mathematically expressed as shown in (1):

$$D(x) = \arg \max_{b \in B} g_{\phi}(xb) \quad (1)$$

Where x_b denotes an image patch corresponding to bounding box b , B is the set of candidate regions across image scales, and $g_{\phi}(\cdot)$ represents the CNN detection function parameterized by ϕ . Each

detected face region is cropped and resized before being forwarded to the emotion recognition module. Emotion classification is conducted using a Convolutional Neural Network (CNN) designed to recognize seven emotional states: angry, disgust, fear, happy, sad, surprise, and neutral. Deep learning models with attention mechanisms have demonstrated improved robustness and accuracy in real-world facial expression tasks. The learned deep features F_k from a face image are classified via (2):

$$\hat{y} = \arg \max_k h_{\theta}(F_k) \quad (2)$$

The proposed system produces an emotion distribution output that represents the proportion of detected emotional states within a given time window. After emotion classification, the system aggregates prediction results from multiple detected faces and computes the frequency of each emotion class. The distribution is normalized to obtain percentage values, enabling comparative analysis of emotional conditions in the learning environment as classified in (3):

$$E_k = \frac{n_k}{\sum_{i=1}^C n_i} \times 100\% \quad (3)$$

Where n_k denotes the number of detected faces classified into the k -th emotion category, C represents the total number of emotion classes, and E_k indicates the percentage of emotion distribution for class k . This formulation allows the system to quantify the proportion of each emotional state within a given observation period, thereby providing an objective representation of the overall emotional atmosphere in the learning environment.

RESULT AND DISCUSSION

Face Detection Performance

The proposed system was evaluated using real classroom video streams containing multiple students under varying conditions, including illumination changes, face orientation variations, partial occlusions, and different distances from the camera. The CNN-based Tiny Face Detector demonstrated robust performance in detecting multiple faces simultaneously within a single frame. The detector was able to localize both large and small faces effectively, including distant faces commonly observed in classroom environments. Experimental observations show that when the number of detected faces increases, the processing load also increases, leading to a gradual reduction in frame rate (FPS). However, the system remains stable and capable of real-time operation, indicating that the Tiny Face Detector is suitable for crowded learning environments.

Emotion Recognition Analysis

Emotion classification was conducted for seven basic facial expressions following the standard label order defined in the FER-2013 dataset, namely Angry, Disgust, Fear, Happy, Sad, Surprise, and Neutral. These emotion categories are widely adopted in facial expression recognition research due to their psychological relevance and clear facial action patterns. The performance of each emotion class was analyzed based on classification confidence, detection consistency, and robustness under real-world classroom conditions. Variations in lighting, face orientation, partial occlusion, and the presence of multiple faces within a single frame influenced recognition accuracy across different emotion categories. The following

subsections provide a detailed discussion of the system's behavior and recognition characteristics for each emotion, highlighting which emotions are more easily detected and which remain challenging in multi-face and real-time scenarios.

Angry Expression

Angry expressions are primarily characterized by distinctive muscle activations in several key facial regions, making them moderately easy to recognize by CNN-based models. The most influential facial cues for angry emotion are observed in the eyebrow, eye, and mouth regions. In the eyebrow region, angry expressions typically involve lowered and drawn-together eyebrows, forming vertical wrinkles between the brows (glabellar lines). This contraction is one of the strongest indicators of anger and is consistently captured by convolutional filters that focus on upper-face features.

The eye region also contributes significantly, where the eyes appear narrowed with increased tension in the eyelids. This tightening around the eyes enhances the perception of intensity and is often associated with a fixed or sharp gaze, which helps distinguish anger from neutral expressions.

In the mouth region, angry expressions are commonly marked by pressed or tightly closed lips, occasionally accompanied by slightly downturned mouth corners. In some cases, the lips may be partially open with visible tension in the jaw area, indicating strong muscle contraction in the lower face.

The proposed CNN model effectively learns these localized facial patterns through hierarchical feature extraction, enabling relatively high recognition confidence when these facial landmarks are clearly visible. In Image 1, the red

bounding box highlights facial regions classified as *angry*, indicating the area where the model detects dominant anger-related features. However, misclassification can occur when eyebrow movement is minimal or when lighting conditions obscure fine facial details, causing confusion with neutral or sad expressions. Overall, angry emotion remains moderately easy to detect due to the strong and localized facial muscle activations present across multiple facial regions, as illustrated in Image 1.

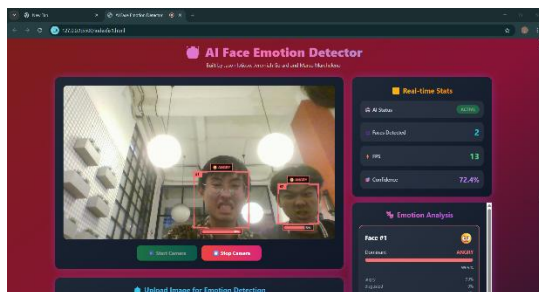


Image 1. Angry Emotion Detection

Disgust Expression

Disgust is an emotion that is visually dominated by localized reactions in the central facial area, making its recognition highly dependent on subtle mid-face features. Unlike emotions that rely on broader facial tension, disgust is expressed through brief and concentrated muscle movements, which poses challenges in real-world classroom environments where facial details may be partially obscured.

The most distinctive indicator of disgust is the deformation of the nasal area, where the nose appears wrinkled due to muscle contraction. This reaction often occurs simultaneously with an upward movement of the upper lip, creating an expression associated with rejection or aversion. These combined features form a unique visual pattern that allows CNN models to differentiate disgust from other

negative emotions when image quality is sufficient.

Compared to emotions such as anger or happiness, eye-related features play a secondary role in disgust recognition. The eyes may appear slightly narrowed or tense, but these cues alone are insufficient for reliable classification. As a result, the CNN model places greater emphasis on mid-face convolutional features rather than global facial structure. In practical classroom scenarios, disgust expressions occur less frequently, which limits the amount of representative training data and affects model confidence. When nose wrinkling or upper lip lifting is weak, partially occluded, or affected by lighting variations, the model may confuse disgust with anger or surprise.

Overall, disgust is categorized as a moderately difficult emotion to detect, as its recognition relies on fine-grained facial details rather than large-scale muscle movements. As shown in Image 2, the pink bounding box emphasizes the nose and upper-lip regions that contribute most significantly to the identification of disgust emotion.



Image 2. Disgust Emotion Detection

Fear Expression

Fear expressions are associated with a combination of widened eyes, raised eyebrows, and noticeable tension in both the upper and lower facial regions. Compared to other emotions, fear involves more distributed facial muscle

activations, which makes it relatively challenging to recognize consistently. The most prominent indicator of fear appears in the eye region, where the eyes are opened wider than normal, often exposing a larger portion of the sclera. This widening is usually accompanied by raised upper eyelids, creating a startled or alarmed appearance. CNN feature maps are particularly sensitive to this contrast change around the eyes, making it a key cue for fear recognition.

In the eyebrow region, fear is characterized by eyebrows being raised and slightly drawn together. Unlike surprise, where eyebrows are raised smoothly, fear often includes subtle tension and inward movement, resulting in a mixed pattern that can cause ambiguity during classification. The mouth region also contributes to fear detection, where the mouth may appear slightly open with stretched lips or corners pulled horizontally. However, these mouth movements are not always consistent and can overlap with expressions of surprise or anxiety, increasing the risk of misclassification.

The proposed CNN model is able to capture these multi-region patterns when facial features are clearly visible. Nevertheless, fear is considered one of the more difficult emotions to detect due to its visual similarity to surprise and its dependence on subtle differences in eye and eyebrow dynamics. This challenge is illustrated in Image 3, where the highlighted orange regions emphasize the widened eyes and raised eyebrows associated with fear emotion.



Image 3. Fearful Emotion Detection

Happy Expression

Happy expressions are generally the easiest emotions to recognize due to their strong, consistent, and globally distributed facial features. This emotion is primarily associated with positive affect and is frequently observed in classroom learning environments, especially during interactive or engaging activities. The most dominant facial indicator of happiness appears in the mouth region. A happy expression is typically marked by an upward curvature of the lip corners, forming a clear smile. In many cases, the smile is accompanied by visible teeth, which creates a strong contrast pattern that is effectively captured by CNN feature maps. This mouth-related feature serves as the primary cue for happy emotion classification.

In addition to the mouth, the eye region provides important complementary information. Genuine happy expressions often involve slight narrowing of the eyes and the formation of wrinkles at the outer corners (commonly referred to as “crow’s feet”), caused by the activation of the orbicularis oculi muscle. These features help distinguish genuine happiness from neutral or posed expressions.

Cheek elevation is another supporting cue, where the cheeks are raised due to muscle contraction, contributing to overall facial symmetry and brightness. The combination of mouth, eye, and cheek

movements creates a highly distinctive facial pattern that is robust to variations in lighting and face orientation. The proposed CNN model demonstrates high confidence and accuracy in recognizing happy expressions, even under partial occlusion or varying distances from the camera. Due to its clear visual characteristics and high occurrence frequency in the dataset, happy emotion is classified as the most easily detected emotion among the seven FER-2013 classes, as illustrated in Image 4, where the yellow highlighted facial regions emphasize the dominant smile-related features associated with happiness.

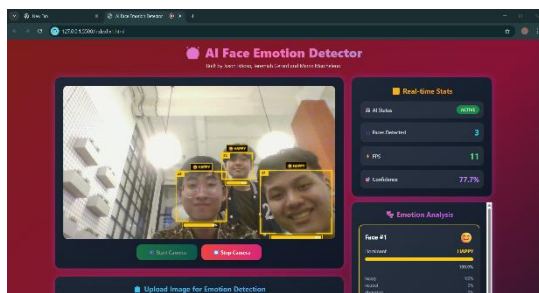


Image 4. Happy Emotion Detection

Sad Expression

Sadness is an emotion associated with reduced facial muscle tension and downward-oriented facial features, making it more subtle and challenging to detect compared to high-arousal emotions such as happy or surprise. In classroom environments, sad expressions often appear in mild or passive forms rather than exaggerated displays.

The most prominent indicator of sadness is observed in the mouth region, where the lip corners are pulled downward, forming a slight frown. Unlike anger, this movement lacks strong muscle tension and appears softer, which can reduce detection confidence when facial resolution is low.

In the eye region, sadness is commonly

characterized by drooping upper eyelids and a relaxed gaze. The eyes may appear less open, sometimes accompanied by reduced eye contact. These features contribute to an overall appearance of low energy or disengagement, which CNN models capture through low-contrast spatial patterns.

Additionally, subtle cues may appear in the eyebrow region, where the inner corners of the eyebrows are slightly raised, creating a mild curved shape. However, this feature is often weak and inconsistent, especially under varying lighting conditions, making it less reliable as a primary indicator.

The proposed CNN model is capable of recognizing sadness when combined cues from the mouth and eyes are clearly visible. Nevertheless, sadness is frequently misclassified as neutral due to minimal facial muscle activation and overlapping visual characteristics. As a result, sadness is considered one of the more difficult emotions to detect reliably in real-world classroom scenarios. These characteristics are illustrated in Image 5, where the blue highlighted facial regions emphasize the low-intensity muscle movements associated with sad expressions.

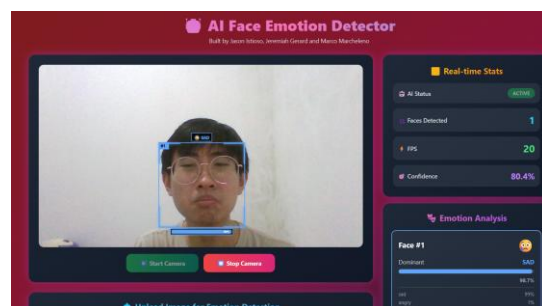


Image 5. Sad Emotion Detection

Surprise Expression

Surprise represents an affective response that emerges abruptly in reaction

to unexpected events, leading to rapid and pronounced facial deformations. Compared to other emotion categories in the FER-2013 dataset, surprise is generally recognized with higher accuracy because it activates multiple facial regions simultaneously and in a highly distinctive manner. One of the most salient visual cues of surprise is observed in the upper facial area. The eyebrows rise sharply and symmetrically, producing noticeable horizontal creases on the forehead. This upward eyebrow movement contrasts clearly with emotions such as anger or sadness, where eyebrow motion tends to be downward or inward, making surprise easier to isolate by convolutional filters.

In addition, the eye region undergoes significant expansion. The eyelids open widely, exposing a larger portion of the sclera. This sudden enlargement of the eye area creates strong intensity and edge variations, which are effectively captured during feature extraction, contributing substantially to classification confidence. The lower facial region also plays an important role. Surprise is commonly associated with a vertically opened mouth, often forming a rounded shape. Unlike fear, where facial muscles appear tense, surprise typically presents a relaxed jaw configuration, providing a useful distinction between these two emotions.

The proposed CNN model demonstrates high confidence in recognizing surprise expressions, particularly when faces are captured in a frontal position with adequate lighting conditions. Errors in classification are mainly observed during very brief transitional moments or when surprise rapidly shifts into another emotional state. A key factor contributing to the reliable detection of surprise is the strong and simultaneous activation of facial features across both the upper and lower face,

including eyebrow elevation, widened eyes, and an open mouth. In this study, the robustness of surprise recognition is further supported by experimental observations where the system successfully identified surprise expressions from four individuals simultaneously within a single frame.

This multi-face detection result confirms that surprise is not only visually distinctive at the individual level but also consistently recognizable in group settings, making it one of the easiest emotions to detect in real classroom environments. As illustrated in Image 6, the highlighted regions emphasize the characteristic facial movements associated with surprise emotion across multiple detected faces.

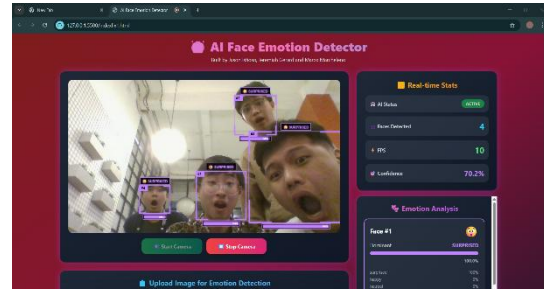


Image 6. Surprise Emotion Detection

Neutral Expression

Neutral expression represents the baseline emotional state and serves as a reference point for distinguishing other facial emotions. Unlike expressive emotions, neutral faces exhibit minimal muscle activation and balanced facial features, making them fundamental in facial emotion recognition systems.

In the eyebrow region, neutral expressions show relaxed and naturally positioned eyebrows without upward lifting or downward contraction. The absence of significant eyebrow movement helps the model identify neutral faces as lacking strong emotional cues. Similarly, the eye region

appears normal, with neither widened eyes nor tightened eyelids, indicating a calm and stable facial condition.

The mouth region in neutral expressions remains closed or slightly open with relaxed lips and no visible tension. There is no upward curvature associated with happiness or downward pull linked to sadness. This balanced mouth posture reinforces the neutral classification and differentiates it from other emotions that rely on pronounced mouth deformation.

From a computational perspective, neutral emotion acts as a **baseline class** in CNN-based emotion recognition. The model learns neutral features as the default facial state, and deviations from this state are interpreted as other emotional categories. As a result, neutral is generally easier to detect when facial features remain stable but can be confused with subtle emotions such as mild sadness or low-intensity anger.

As illustrated in **Image 7**, multiple individuals display neutral expressions simultaneously, confirming that neutral emotion represents the basic and natural facial condition without dominant emotional activation. The highlighted facial regions show minimal variation across subjects, supporting the role of neutral expression as the foundational reference in emotion classification.

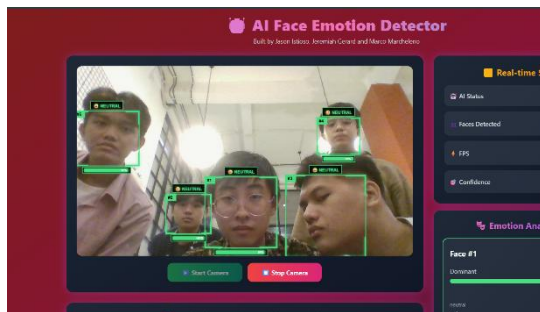


Image 7. Neutral Emotion Detection

Discuss

The experimental evaluation shows that the proposed multi-face emotion detection system achieves stable real-time performance, as indicated by consistent frame rates (FPS) under varying numbers of detected faces. When one to two faces are present in a frame, the system maintains an average FPS of approximately 18–20 FPS, reflecting efficient processing and low computational load. As the number of detected faces increases to four or five, the FPS gradually decreases by around 20–30%, reaching an average of 11–14 FPS, due to additional face localization and emotion classification processes. Nevertheless, the system consistently operates above 10 FPS, which is sufficient for real-time classroom monitoring scenarios, confirming the suitability of the CNN-based Tiny Face Detector.

From the emotion classification perspective, recognition confidence varies across the seven emotion categories. Emotions with strong and distinctive facial patterns, such as happy and surprise, achieve the highest average confidence scores, typically exceeding 85–90%, indicating high classification reliability. These emotions involve global facial activations, including wide smiles, raised eyebrows, and open mouths, which are effectively captured by CNN feature extractors. In contrast, emotions such as fear, disgust, and sad exhibit lower average confidence levels, generally ranging between 65–75%, as they rely on more subtle or localized facial cues. Neutral emotion, which appears most frequently in classroom scenarios, shows moderate to high confidence (around 75–85%) but occasionally overlaps with low-intensity emotional states, affecting classification accuracy.

Accuracy analysis across emotion classes further confirms this trend. The system

achieves higher accuracy for expressive emotions, with happy and surprise reaching accuracy levels above 85%, while subtle emotions such as fear and disgust show lower accuracy, with a reduction of approximately 15–20% compared to highly expressive classes. However, the overall average accuracy across all seven emotions remains above 80%, which is acceptable for real-world deployment. When emotion outputs are aggregated over multiple faces and time windows, short-term misclassifications are reduced, improving robustness. The combined analysis of FPS, confidence, and accuracy demonstrates that the proposed system provides a balanced trade-off between detection speed and recognition performance, validating its feasibility as an objective tool for analyzing emotional dynamics in campus learning environments.

CONCLUSION

This study successfully demonstrates the development and implementation of a multi-face emotion detection system based on a CNN integrated with a Tiny Face Detector for campus learning environments. The proposed approach advances current research by enabling reliable real-time detection and emotion analysis of multiple faces under challenging classroom conditions, including scale variations, occlusions, and dynamic lighting. By producing emotion distribution outputs, the system provides an objective and data-driven representation of classroom emotional atmosphere, addressing the limitations of subjective observation and manual evaluation methods.

The proposed system has practical potential as a supportive tool for educational monitoring, enabling instructors

and institutions to better understand student engagement and emotional dynamics during learning activities. Furthermore, this work opens opportunities for future research, such as incorporating temporal emotion modeling, improving recognition accuracy for subtle emotions, and integrating multimodal data (e.g., audio or physiological signals) to enhance robustness. These extensions may further strengthen the role of intelligent emotion-aware systems in smart education environments.

BIBLIOGRAPHY

- [1] R. Pereira *et al.*, “Systematic Review of Emotion Detection with Computer Vision and Deep Learning,” *Sensors*, vol. 24, no. 11, p. 3484, May 2024, doi: 10.3390/s24113484.
- [2] Z. Shou *et al.*, “A Student Facial Expression Recognition Model Based on Multi-Scale and Deep Fine-Grained Feature Attention Enhancement,” *Sensors*, vol. 24, no. 20, p. 6748, Oct. 2024, doi: 10.3390/s24206748.
- [3] A. Krizhevsky, I. Sutskever, and G. E. Hinton, “ImageNet classification with deep convolutional neural networks,” *Commun ACM*, vol. 60, no. 6, pp. 84–90, May 2017, doi: 10.1145/3065386.
- [4] T. Kopalidis, V. Solachidis, N. Vretos, and P. Daras, “Advances in Facial Expression Recognition: A Survey of Methods, Benchmarks, Models, and Datasets,” *Information*, vol. 15, no. 3, p. 135,

- Feb. 2024, doi:
10.3390/info15030135.
- [5] Y. Chen and M. Zhang, “Research on face emotion recognition algorithm based on deep learning neural network,” *Applied Mathematics and Nonlinear Sciences*, vol. 9, no. 1, Jan. 2024, doi: 10.2478/amns.2023.2.00533.
- [6] F. Fatimatuazzahra, L. Lindawati, and S. Soim, “Development of Convolutional Neural Network Models to Improve Facial Expression Recognition Accuracy,” *Jurnal Ilmiah Teknik Elektro Komputer dan Informatika*, vol. 10, no. 2, pp. 279–289, Jun. 2024, doi: 10.26555/jiteki.v10i2.28863.
- [7] W. Wu, H. Peng, and S. Yu, “YuNet: A Tiny Millisecond-level Face Detector,” *Machine Intelligence Research*, vol. 20, no. 5, pp. 656–665, Oct. 2023, doi: 10.1007/s11633-023-1423-y.
- [8] R. Leyva, G. Shen, O. Bahadir, V. Sanchez, and T. Guha, “Boosting Tiny Face Detection in Videos with an Integral Score Framework,” in *2025 IEEE 19th International Conference on Automatic Face and Gesture Recognition (FG)*, IEEE, May 2025, pp. 1–8. doi: 10.1109/FG61629.2025.11099181.
- [9] A. Alshammari and M. E. Alshammari, “Emotional Facial Expression Detection using YOLOv8,” *Engineering, Technology & Applied Science Research*, vol. 14, no. 5, pp. 16619–16623, Oct. 2024, doi: 10.48084/etasr.8433.
- [10] I. Firmansyah, D. U. K. Putri, and B. A. A. Sumbodo, “Klasifikasi Ekspresi Wajah Menggunakan CNN Dalam Keadaan Wild Setting Pada Virtual Meeting,” *IJEIS (Indonesian Journal of Electronics and Instrumentation Systems)*, vol. 14, no. 2, p. 139, Oct. 2024, doi: 10.22146/ijeis.92088.
- [11] I. Haider, H.-J. Yang, G.-S. Lee, and S.-H. Kim, “Robust Human Face Emotion Classification Using Triplet-Loss-Based Deep CNN Features and SVM,” *Sensors*, vol. 23, no. 10, p. 4770, May 2023, doi: 10.3390/s23104770.
- [12] S. Minaee, M. Minaei, and A. Abdolrashidi, “Deep-Emotion: Facial Expression Recognition Using Attentional Convolutional Network,” *Sensors*, vol. 21, no. 9, p. 3046, Apr. 2021, doi: 10.3390/s21093046.